Pham Quoc Cuong

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Summary

Fresher Software Engineer focused on MES and manufacturing software. Build full-stack applications using C++, Python, React, PostgreSQL and optimize complex production scheduling problems with genetic algorithms. Solid foundation in data structures, algorithms, and database design.

Projects

MES-APS: Production Scheduling System ↗ | *Live Demo* ↗ Jun. 2026 – Oct. 2026
Full-stack developer (individual project) · C++17, Node.js (N-API), React, PostgreSQL

- Built and deployed an MES-style scheduling system that validates CSV/Excel work orders and schedules jobs based on due dates, priorities, setup times, machine efficiency, and downtime, with support for rescheduling after machine breakdowns.
- Built a Gantt dashboard with on-time delivery, tardiness, utilization and bottleneck KPIs, plus run comparison and an audit log for planners.
- Designed a PostgreSQL schema and a Node.js REST API; offloaded C++17 scheduler to an N-API worker to avoid event-loop blocking (0.55 s/run locally); verified with 10 black-box test cases.
- Implemented a genetic algorithm reaching best-known solutions on 16/20 Lawrence benchmarks (best of 50 seeded runs); on a congested 10-order test it had 0 late orders vs. 3 (FIFO) and 2 (EDD) simple baselines.

MindMelody: Emotion-Aware Music Therapy Platform ↗ | *Live Demo* ↗ Nov. 2025 – Jun. 2026
Full-stack developer (graduation thesis) · Django, React, PostgreSQL, Gemini API

- Built a full-stack platform that uses the DASS-21 questionnaire to quantify users' mental states and deliver emotion-aware music and chat interventions.
- Processed 114K+ Spotify track records from a public Kaggle dataset to create a cleaned dataset of 50K+ tracks; developed a 4D weighted euclidean distance algorithm (validated via Monte Carlo simulations) to generate dynamic, mood-shifting playlists (ISO principle).
- Integrated the Gemini API with static RAG into a CBT-style chatbot, with an implicit feedback loop and keyword-based guardrails that route crisis messages to an SOS flow.

Technical Skills

Languages: C++, JavaScript, Python, SQL
Frameworks: React, Tailwind CSS, Django, Node.js/Express
Databases: PostgreSQL, SQL Server, MongoDB
Tools: Git, GitHub, Vercel, Render, Supabase, Jupyter Notebook
Concepts: DSA, REST API design, relational schema design, genetic algorithms, production scheduling

Education

Bachelor of Engineering in Software Engineering Sep. 2022 – Jul. 2026
Quang Ninh University of Industry *GPA: 3.61/4.00*

Experience

Programming Tutor & Competition Mentor Jun. 2025 – Apr. 2026
Self-employed *Quang Ninh, Vietnam*

- Provided one-on-one mentoring in C++ and Data Structures & Algorithms, preparing a student for specialized IT high-school entrance exams.
- Co-developed a logic-based Scratch application that received third prize in a provincial competition.

Certifications

TOEIC Listening & Reading 2026
ETS / IIG Vietnam *Score: 820/990*

HackerRank SQL Certificate ↗ 2026
HackerRank *SQL Intermediate*